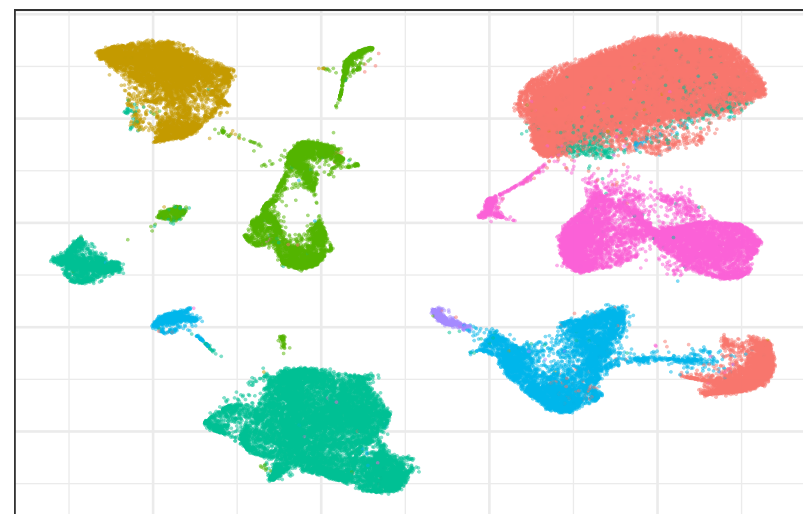


A

Human jaw (GSE303003): 65,454 cells, 4 MRONJ vs 4 cyst, 7 compartments

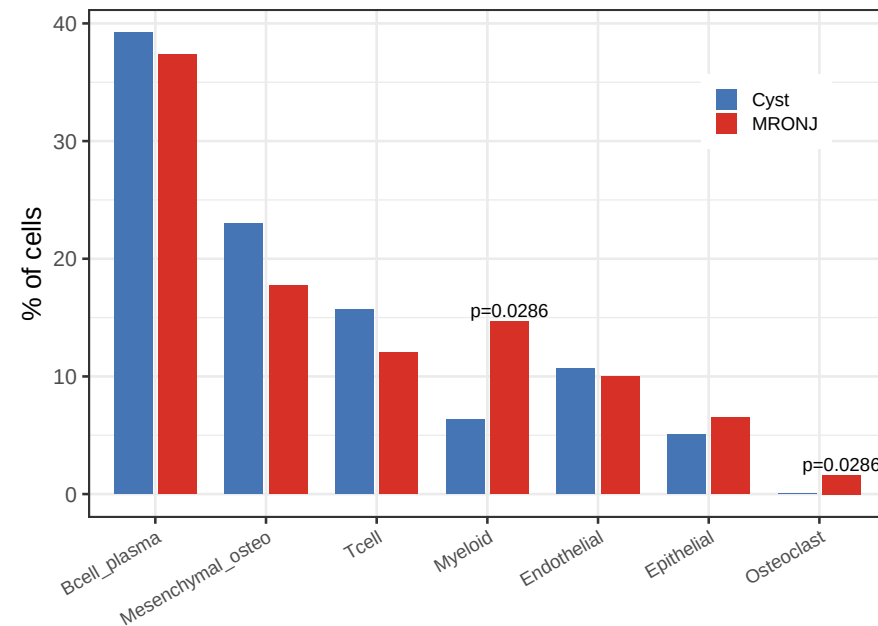


UMAP2

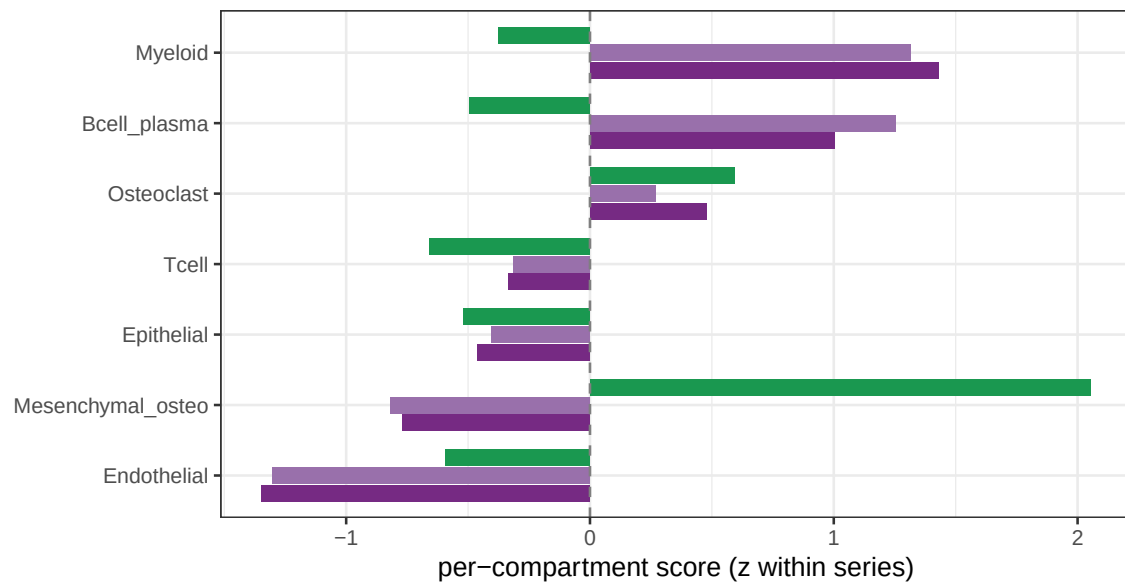
UMAP1

B

Compartment proportions: osteoclast & myeloid expansion in MRONJ (4v4, $p=0.029$)

**C**

Positive control: osteo markers peak in mesenchymal, but the L1 program does NOT
two scorings agree (Spearman $\rho=1$); program highest in myeloid, lowest in mesenchymal/endothelial



per-compartment score (z within series)

■ L1 program (AddModuleScore)
■ L1 program (UCell, rank)
■ osteo markers (RUNX2/SP7/COL1A1) — positive ctrl

D

Program NOT suppressed in MRONJ mesenchyme

whole-program sign test: 294/669 (44%) suppressed-direction, NOT > 50% ($p=1$); Wilcoxon $p=0.057$

